
A
Structural cis-QTL: IR1 Repeat Number Drives EBV CN
Multi-Platform & Artifact Controls:
• 30x IR1 / genome_uniq: r = +0.3148 (p = 1.7e-60)
• 30x IR1 / local_Cp_flank: r = +0.3037 (p = 2.1e-53)
• Within-chrEBV GC-controlled: r = +0.3048 (96.8% retained)
• ONT long-read IR1 vs ONT CN (n=890): r = +0.2945
• MAGE RNA within-pop+batch (n=717): r = +0.3687
30x WGS Within-Pop: r = +0.3148 (p = 1.8e-57, n = 2,449)
Within-Population Demeaned log2 (EBV CN)
Within-Population Demeaned IR1 (BamHI-W) Repeat Units
B
IR1 Expansion Boosts BHRF1 & Lytic Switch vs EBNA-2
WGS log2(EBV CN)
(Episome accumulation)
+0.3687
log2(Lytic IE / Latent)
(Lytic switch ratio)
+0.2531
Lytic IE RPKM
(BZLF1 / BRLF1)
+0.1859
Spliced BZLF1 RPM
(Zta productive switch)
+0.1743
BHRF1 RPKM
(W-spliced viral BCL-2)
+0.1029
EBNA-1 RPKM
(Episome maintenance)
-0.0257
LMP-1 RPKM
(Latent III NF-kB)
-0.0993
EBNA-2 RPKM
(Downstream of IR1)
-0.1068
p = 0.05 threshold (|r| = 0.073)
Within-Pop+Batch Correlation (r) with IR1 Repeat Copy Number (n = 717)
C
Neutrality of IBS/GBR Barcodes vs Replicated Viral QTLs
12,129:T>A (W0/Cp-Wp Leader)
[All 1kGP n=2,449, carrier=111]
+0.170 (r)
12,129:T>A (W0/Cp-Wp Leader)
[Holdout Replication n=1,732]
+0.174 (r)
12,129:T>A -> MAGE Lytic IE
[Within-pop+batch n=717]
+0.105 (r)
Rare Zp Burden (BZLF1 Promoter)
[90,944–91,500 bp, n=16 carriers]
+0.231 Δlog2
91,192:G>A (BRLF1 IBS Barcode)
[Within-pop n=2,449: NEUTRAL]
-0.041 (r)
132,965:G>GC (BXRF1 GBR Barcode)
[Within-pop n=2,449: NEUTRAL]
-0.042 (r)
163,858:G>T (BALF2 ssDNA-BP)
[cis-BALF2 RNA within-pop+batch]
-0.124 (r)
163,858:G>T (BALF2 ssDNA-BP)
[trans-Lytic IE within-pop+batch]
-0.128 (r)
Rare Rp Burden (BRLF1 Promoter)
[92,896–93,500 bp, n=12 carriers]
-0.493 Δlog2
Within-Population Effect Size (Correlation r or Rare Promoter Δlog2[CN])
D
Within-Sample Heteroplasmic Viral cis-ASE
Heteroplasmic SNV observations (n = 2,093, r = +0.9428)
Balanced cis-transcription (VAF_RNA = VAF_DNA)
91,192:G>A (BRLF1 IBS)
20.5% RNA vs 19.2% DNA (neutral)
168[illegible]63:T>G (LMP-1)
6.1% RNA vs 3.9% DNA (p=1.5e-95, cis-UP)
36,470:T>C (EBNA-2)
2.7% RNA vs 16.0% DNA (p=1.1e-31, cis-DOWN)
Same-Donor MAGE RNA Variant Allele Fraction (VAF_RNA)
Within-Sample WGS DNA Variant Allele Fraction (VAF_DNA)